Part 10

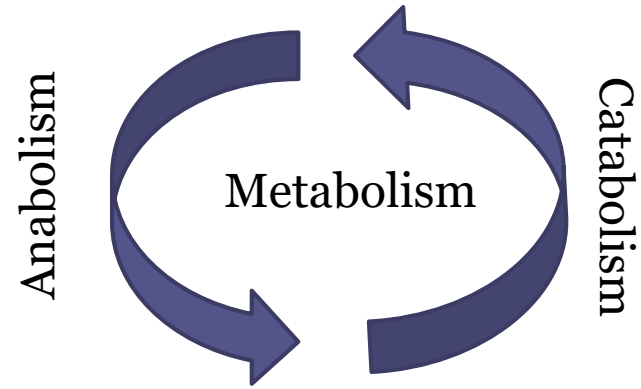
Cell metabolism

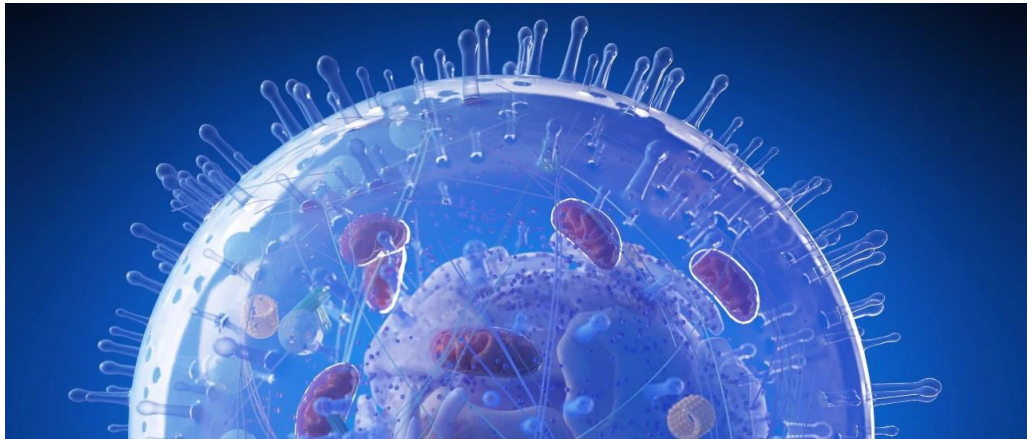


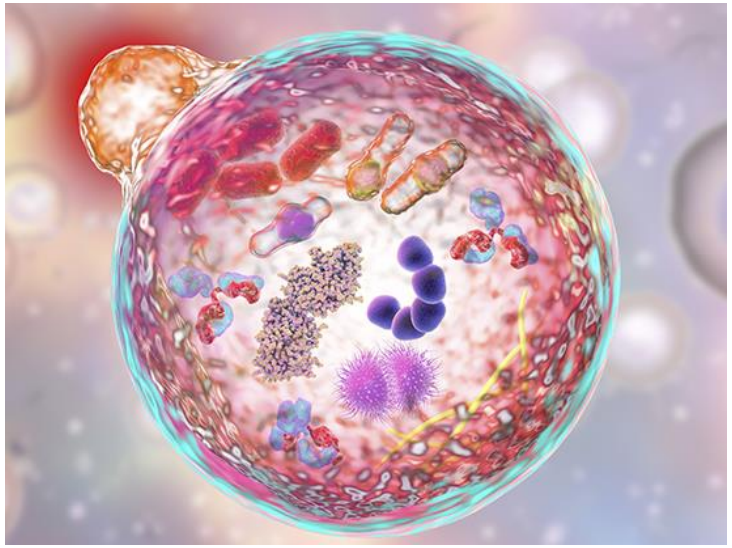
Cellular metabolism is comprised of the chemical reactions that occur in living cells

Anabolism: The synthesis of larger molecules from simpler ones (typically input of energy)

Catabolism: The breakdown of larger molecules into simpler ones (convert nutrients to energy – typically release energy)









Key terms

Exergonic reaction: A reaction that releases energy

Endergonic reaction: A reaction that consumes energy

ATP: Adenosine triphosphate

Enzymes

NAD: Nicotinamide adenine dinucleotide

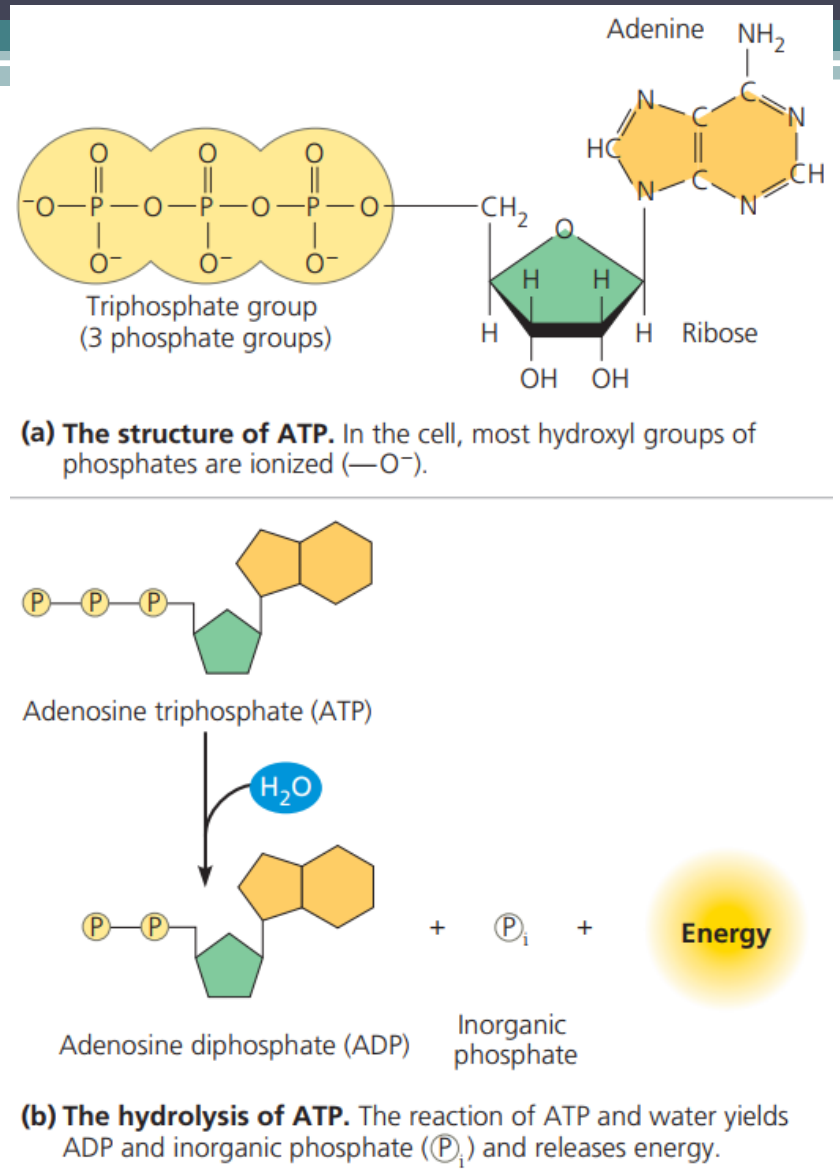
NADP: Nicotinamide adenine dinucleotide phosphate

ATP

Components: sugar ribose, nitrogenous base adenine, a chain of three phosphate groups

The hydrolysis of ATP releases free energy to perform the three types of cellular work- chemical, transport, and mechanical

ATP power cellular work by coupling exergonic reactions to endergonic reactions

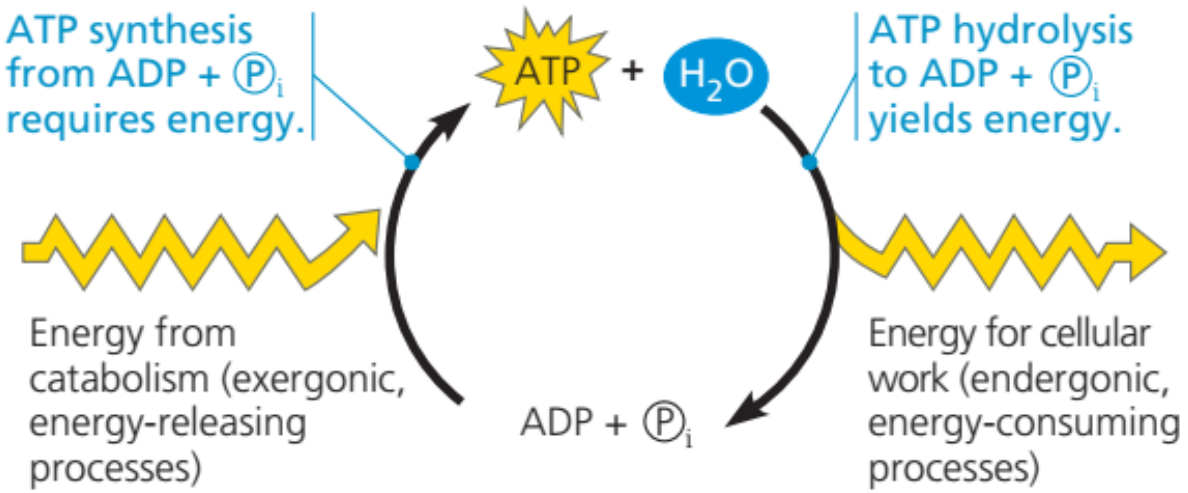


The structure and hydrolysis of adenosine triphosphate (ATP)

ATP regeneration

ATP is a renewable resource that can be regenerated by the addition of phosphate to ADP

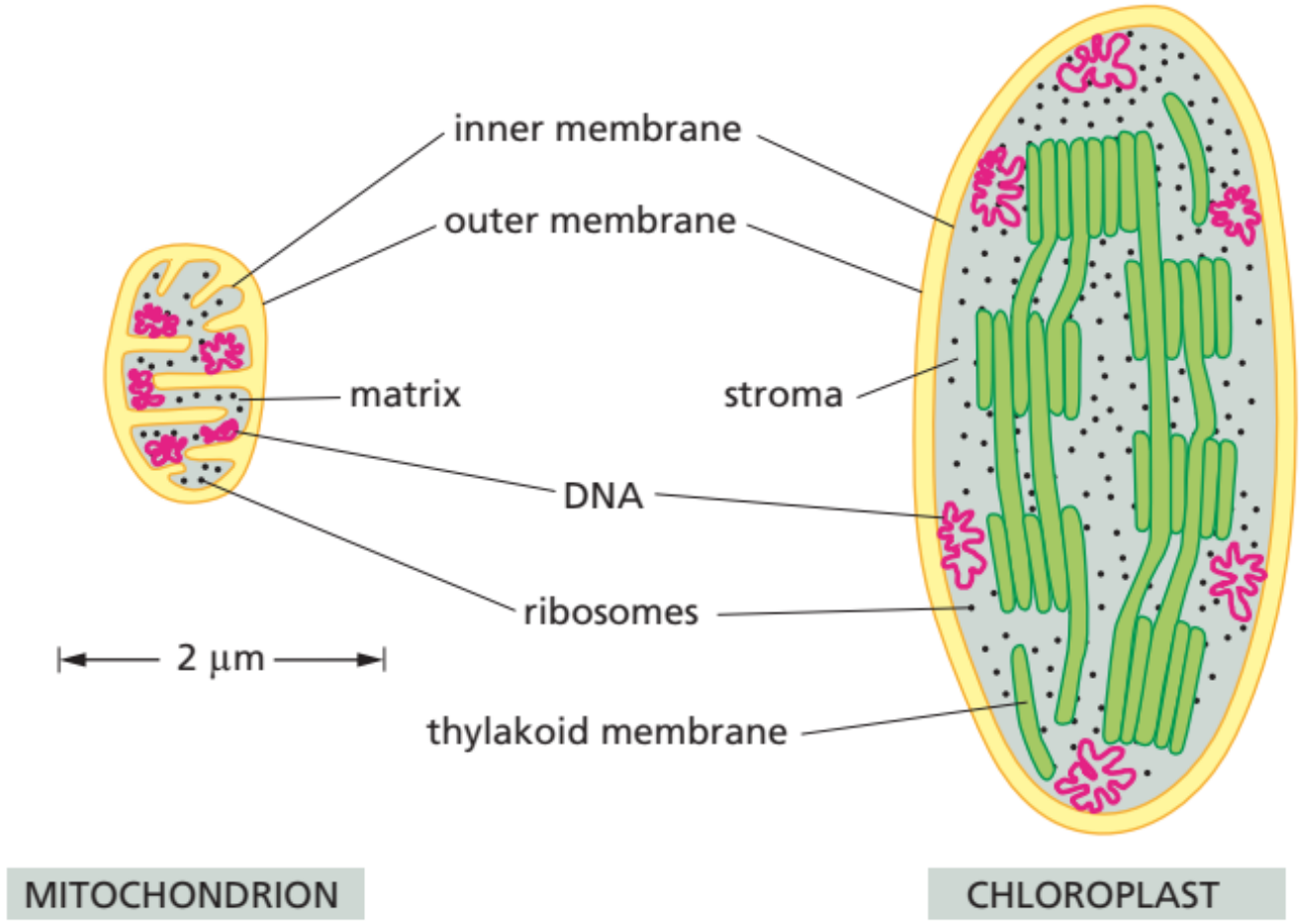
Energy released by breakdown reactions (catabolism) in the cell is used to phosphorylate ADP, regenerating ATP



The ATP cycle

The ATP cycle proceeds at an astonishing pace

Ex: a working muscle cell recycles its entire pool of ATP in less than a minute. That turnover represents 10 million molecules of ATP consumed and regenerated per second per cell



Mitochondria and chloroplasts share many of the features of their bacterial ancestors

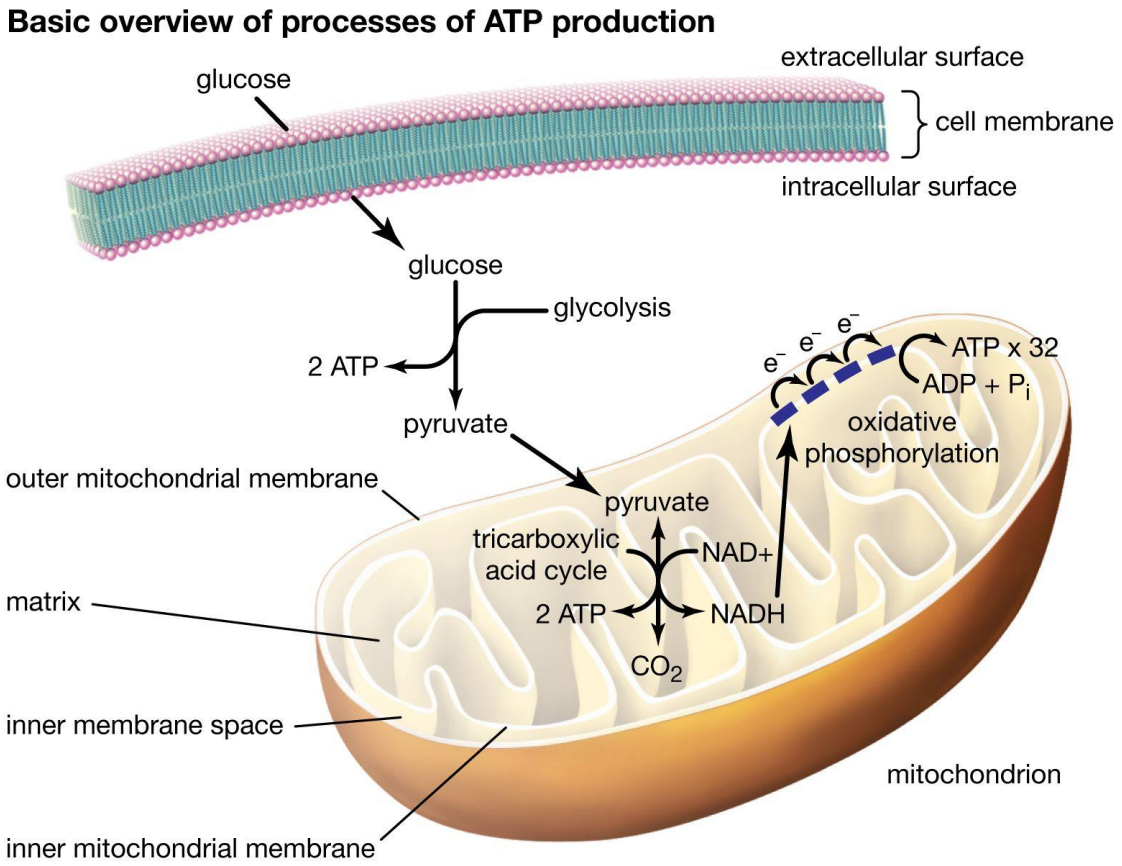
Cellular respiration

$$C_6H_{12}O_6 + 6O_2 \xrightarrow{\text{Cellular respiration}} 6CO_2 + 6H_2O + ATP$$

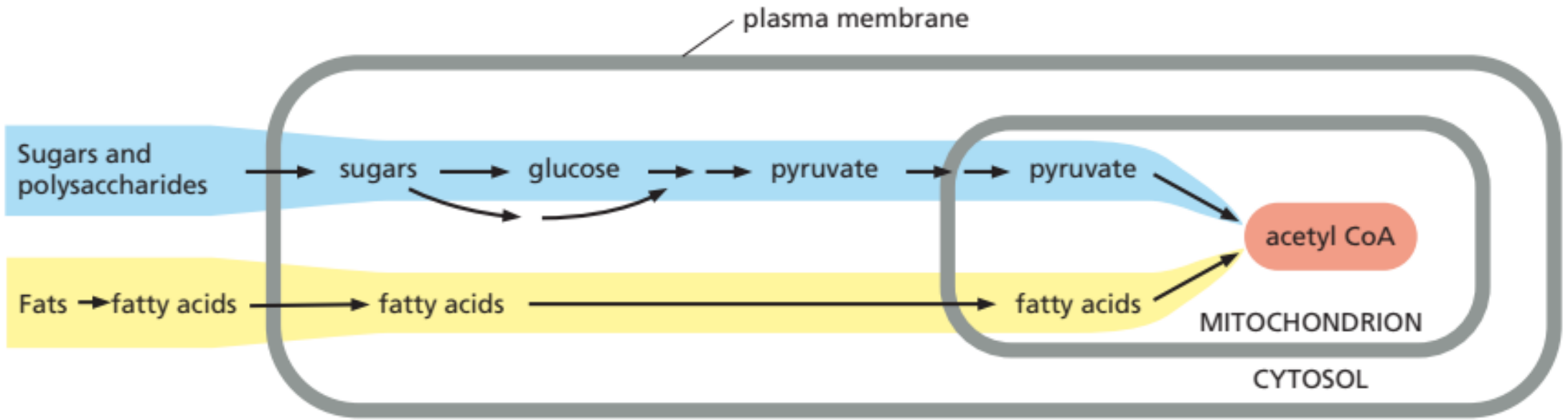
The process by which organisms combine oxygen with foodstuff molecules, diverting the chemical energy and discarding, as waste products, carbon dioxide and water.

Organisms that do not depend on oxygen degrade foodstuffs in a process called fermentation

Three main steps: Glycolysis, the tricarboxylic acid cycle, and oxidative phosphorylation



Cellular respiration



Glycolysis

Glucose + 2 NAD+ + 2 ADP + 2 Pi --> 2 Pyruvate + 2 NADH + 2 H+ + 2 ATP + 2 H2O

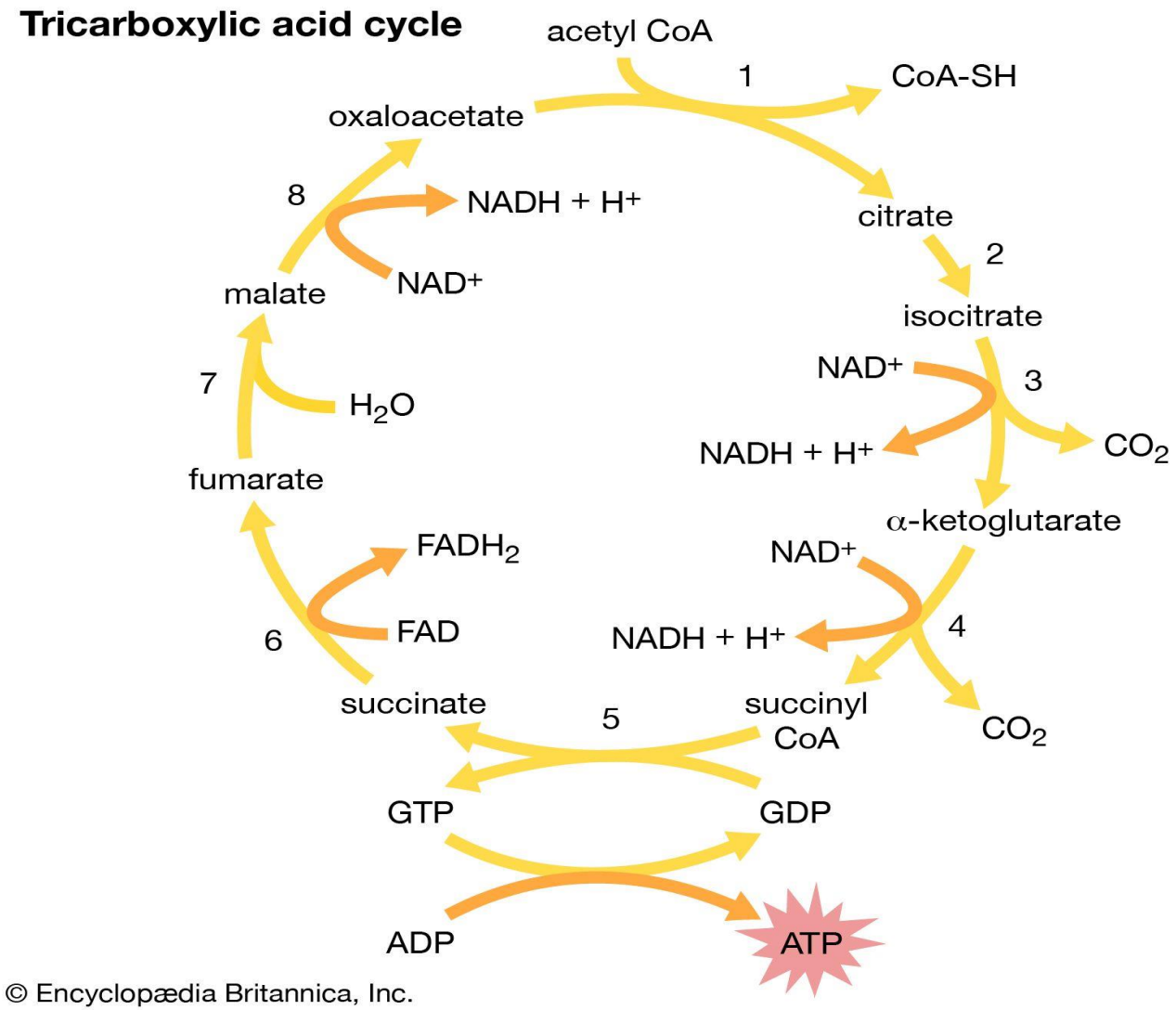
Acetyl CoA is produced in the mitochondria from molecules derived from sugars and fats

Cellular respiration

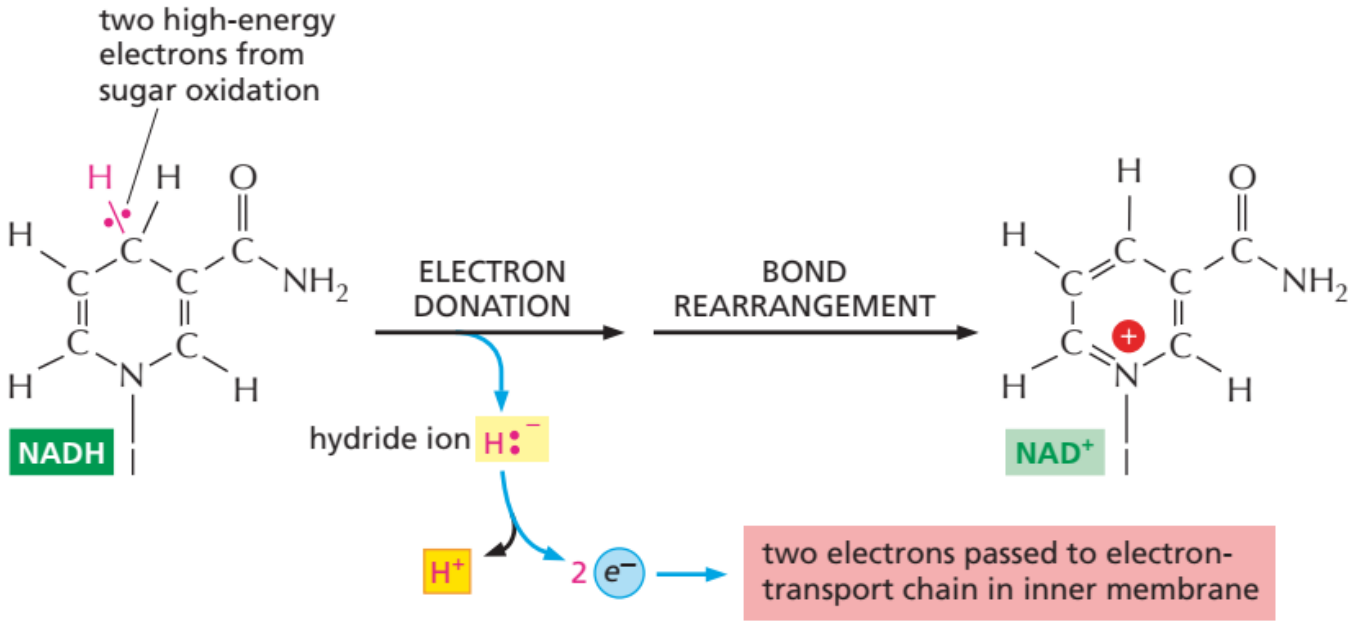
The tricarboxylic acid cycle (TCA, also known as the Krebs cycle or the citric acid cycle) is a series of chemical reactions used in aerobic organisms (pro- and eukaryotes) to generate energy via the oxidation of acetyl-coenzyme A (CoA)

Products of the one turn of the cycle: 01 GTP (or ATP), 03 NADH, 01 FADH2, and 02 CO2

One glucose molecule = 2 acetyl CoA = 2 turns



Cellular respiration



NADH donates its high-energy electrons to the electron transport chain. In this drawing, the electrons being transferred are shown as two red dots on a red hydrogen atom. A hydride ion (a hydrogen atom with an extra electron) is removed from NADH and is converted into a proton and two electrons.

Electrons are also carried in a similar way by FADH2

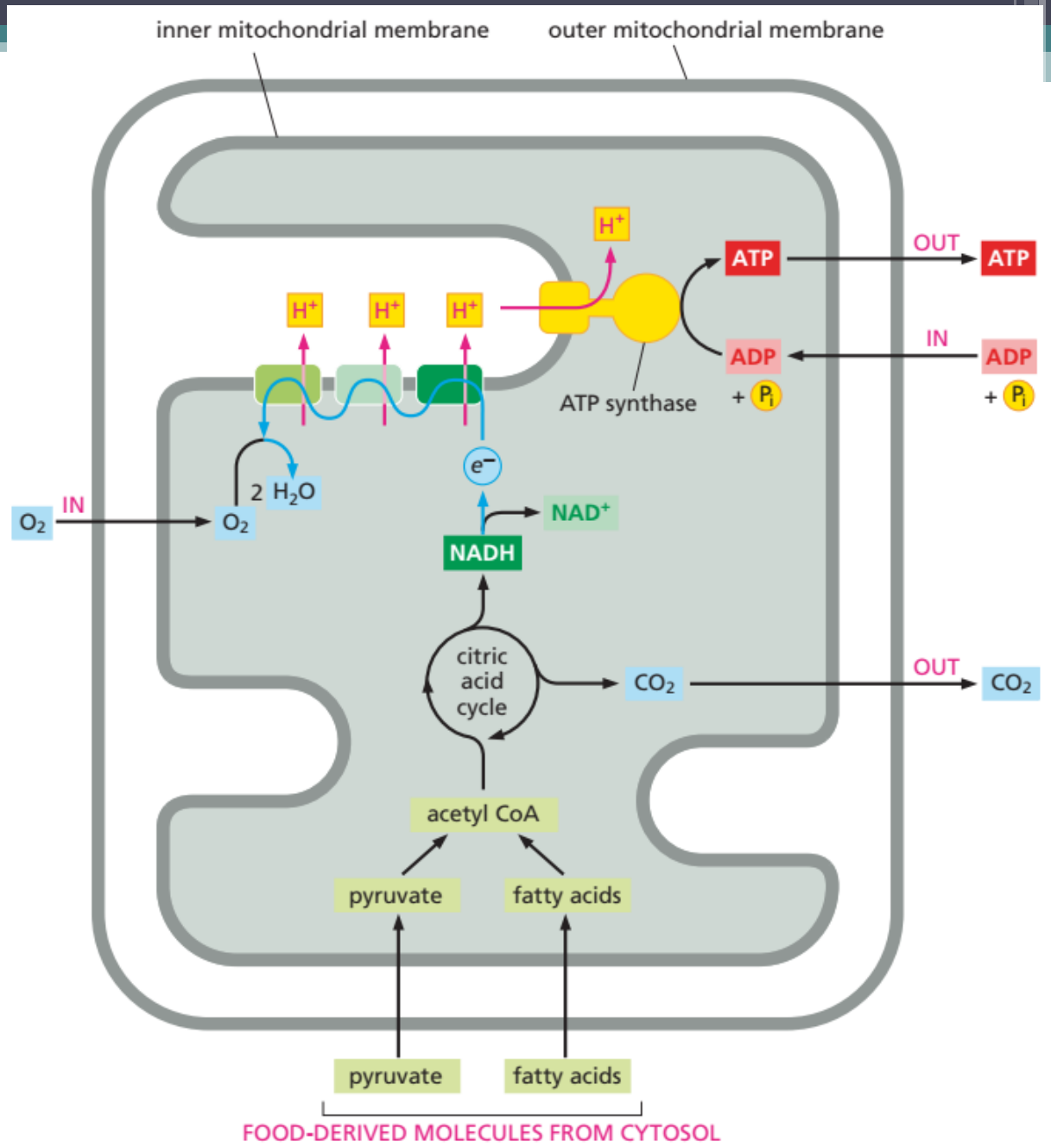
Cellular respiration

Activated carriers generated during the citric acid cycle power the production of ATP.

During oxidative phosphorylation, high energy electrons donated by NADH (and FADH2) are then passed along the electron transport chain in the inner membrane to oxygen (O2); this electron transport generates a proton gradient across the inner membrane, which is used to drive the production of ATP by ATP synthase.

The exact ratios of “reactants” and “products” are not indicated in this diagram: for example, we will see shortly that it requires four electrons from four NADH molecules to convert O2 to two H2O molecules.

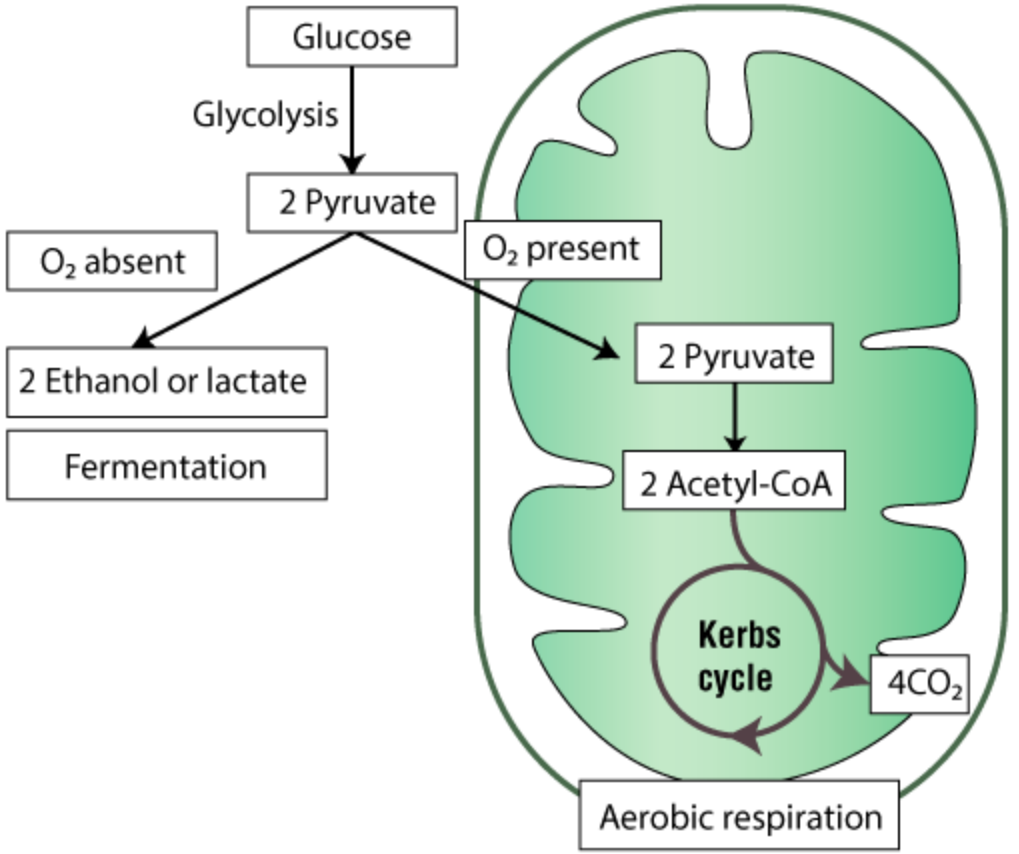
Oxidative phosphorylation produces **26–28** ATP molecules from the Kreb's cycle from one molecule of glucose converted into pyruvate



Cellular respiration

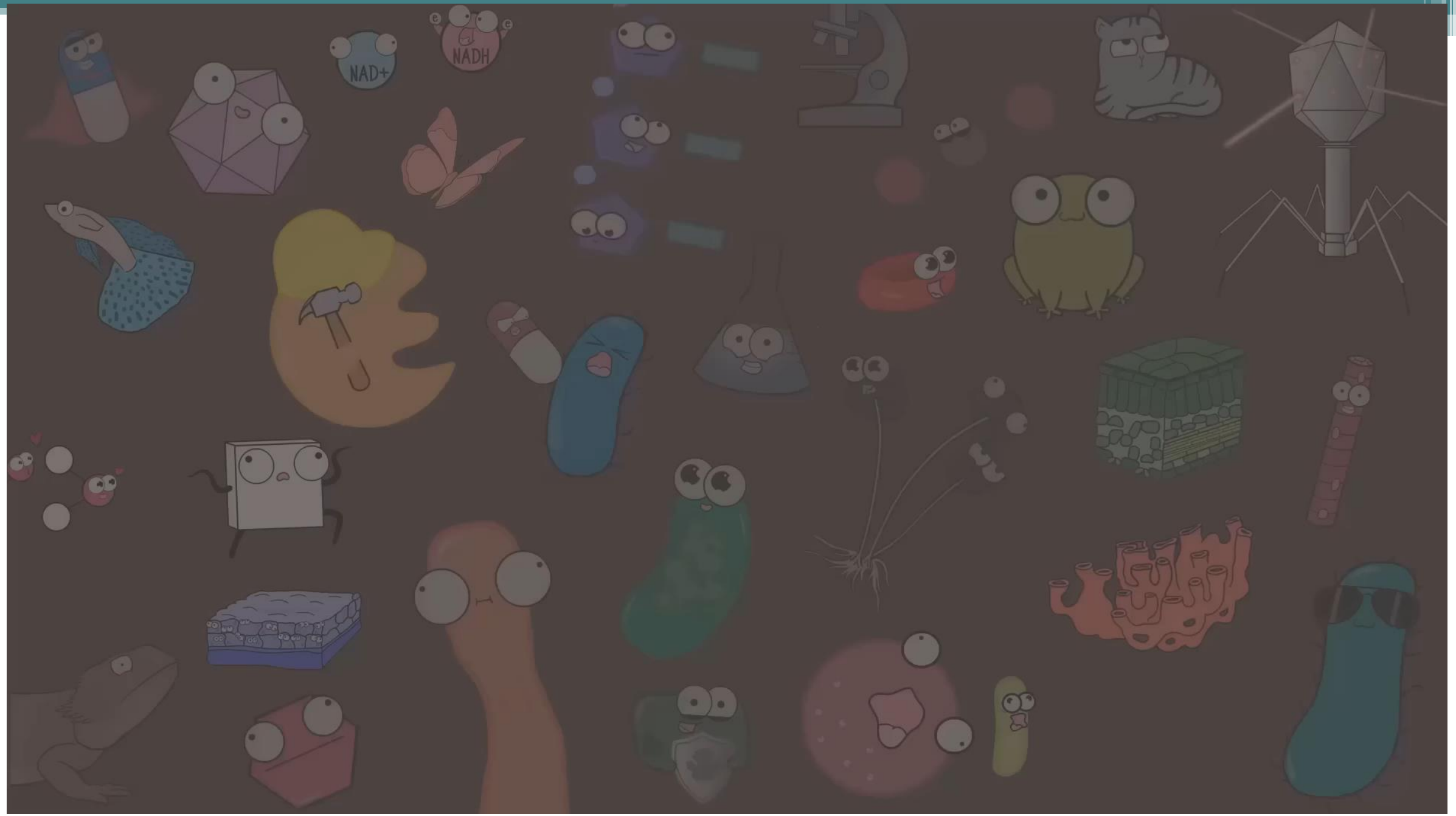
Steps	Glycolysis	Krebs	oxidative phosphorylation
Location	Cytoplasm	Matrix	Inner membrane
Materials	One glucose molecule	2 pyruvate	8 NADH, 2 $FADH_2$, O_2
Products	2 pyruvates, 2 NADH, 2 ATP	$4CO_2$, 6 NADH, 2 $FADH_2$, 2 ATP	26-28 ATP, H_2O

Fermentation

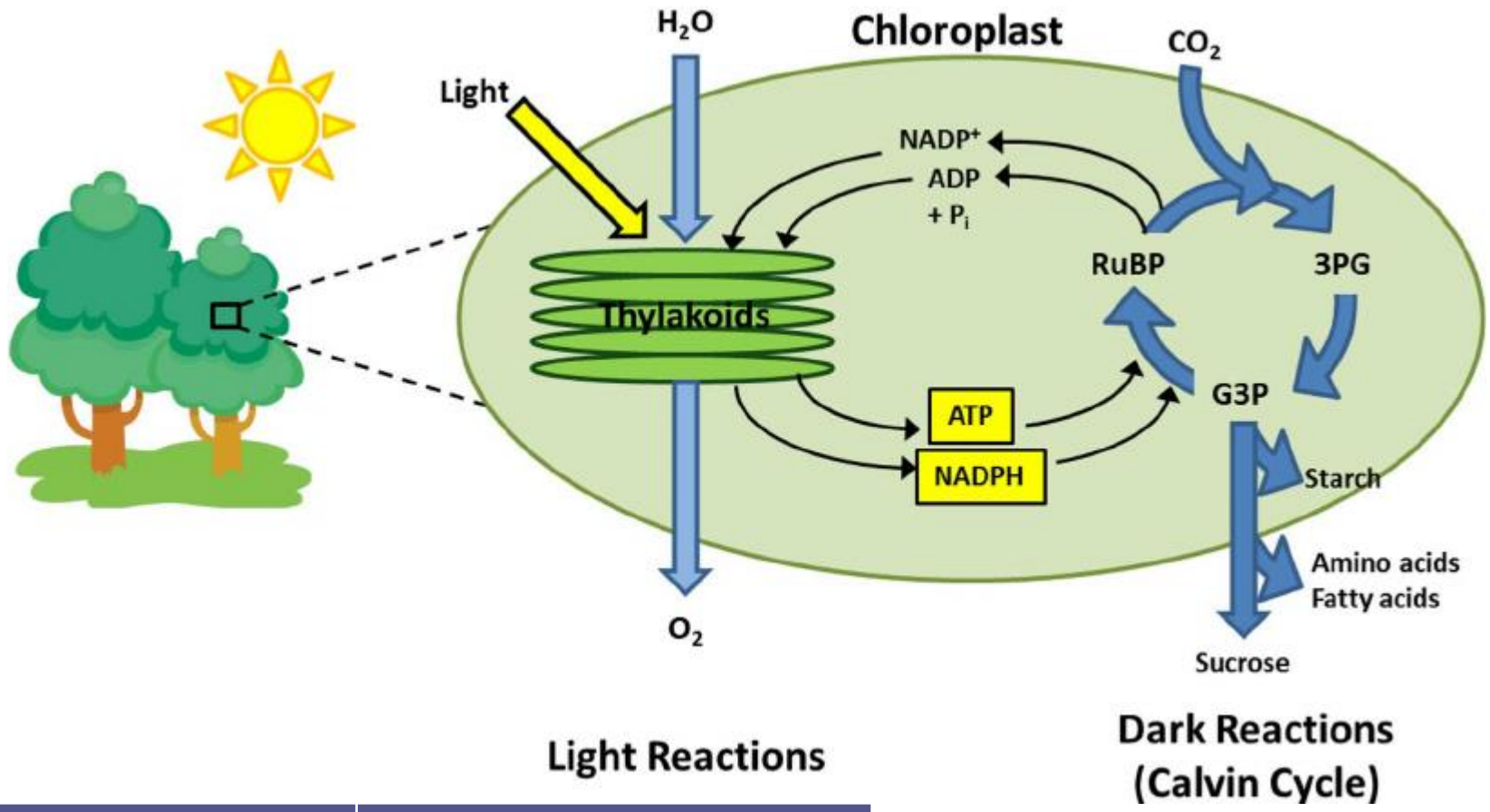


Source: Byjus.com

video



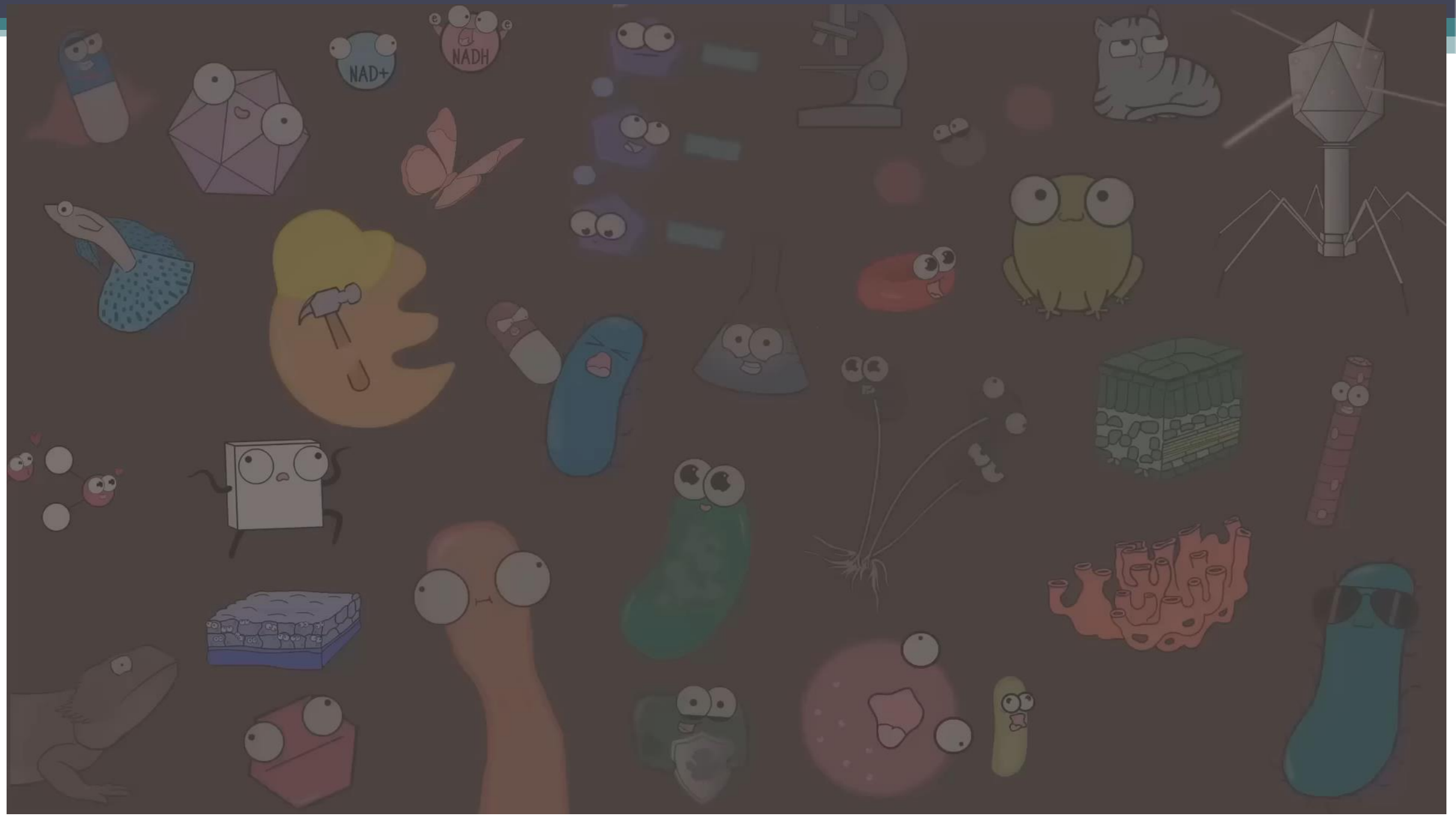
Photosynthesis



Steps	Light reactions	Dark reactions
Locations	Thylakoid	Stroma
Materials	H_2O, light energy	CO_2, ATP, NADPH
Products	O_2. ATP, NADPH	Glucose

Videos

Photosynthesis


NAD+
NADH